

Using Ecological Inference Point Estimates as Dependent Variables in Second-Stage Linear Regressions

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The practice of using point estimates produced by the King ecological inference technique as dependent variables in second-stage linear regressions leads to second-stage results that, in general, are inconsistent. This conclusion holds even when all assumptions behind King's ecological technique are satisfied. Second-stage inconsistency is a consequence of the fact that King-based point estimates of disaggregated quantities contain errors correlated with the true quantities the estimates measure. Our findings on second-stage inconsistency, as well as a fix that we propose, follow from econometric theory in conjunction with an analysis of simulated and real ecological data sets.

1 Introduction

Ecological inference is one of the classic statistical problems in political science (e.g., Robinson 1950; Achen and Shively 1995). Recently, King (1997) proposed an ecological inference technique that produces point estimates of disaggregated quantities that can be used as dependent variables in second-stage regressions. King's technique combined with such regressions can be found, for instance, in Burden and Kimball's (1998) prominent study of ticket splitting. In the first stage of their analysis, Burden and Kimball apply King's ecological inference technique to district-level vote totals for congressional candidates and thereby generate disaggregated estimates of ticket splitting rates. In their second stage, Burden and Kimball use these estimated rates as dependent variables in regressions that analyze why voters cast split-ticket ballots.

The practice of using King-based ecological inference point estimates in second-stage regressions (EI-R) is growing in popularity (Kimball and Burden 1998; Voss and Lublin 1998, 2001; Cohen et al. 1999; Hill 1999; Wolbrecht and Corder 1999; Benoit et al. 2000; Lublin and Voss 2002; Voss and Miller 2000; Gay 2001; Kousser 2002; Burden and Kimball 2002). Furthermore, EI-R has been advocated by King (1997, pp. 279, 290), and given

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the overall level of attention received by King's ecological inference technique, there is every reason to think that EI-R will soon be a standard component of the political science methodological toolbox.

However, we argue that, except in highly unusual circumstances, the second-stage coefficient estimates produced by EI-R are inconsistent. Notably, this conclusion holds even when all assumptions behind King's ecological inference technique are satisfied. The root of the inconsistency that affects EI-R lies in the procedure's first stage, where King's ecological inference technique is used to generate point estimates (hereafter called *EI estimates*) of disaggregated quantities such as district-level rates of ticket splitting. These point estimates are then used as dependent variables in a second-stage regression. Second-stage inconsistency is a result of the fact that EI estimates themselves are contaminated with error, and importantly, the error in individual EI estimates does not tend to zero as the number of observations in an ecological data set goes to infinity. Thus, regardless of the underlying sample size and regardless of whether one uses King-based EI estimates or even EI estimates based on the linear approximation proposed by McCue (2001), EI-based second-stage regressions have a dependent variable that is measured with error.

It is well known that having a dependent variable measured with an error that is uncorrelated with the variable's true value does not cause inconsistency in a regression's slope parameter estimates. Instead, it leads only to inefficiency and, at worst, an inconsistent estimate of the regression's constant term (Greene 1997, p. 436). However, if the error affecting a dependent variable is actually correlated with it, then all regression estimates based on the contaminated variable are at risk of inconsistency.

Indeed, we show that this risk is realized in the case of EI-R because errors in EI estimates are correlated with the true disaggregated quantities that the estimates measure. In particular, King's ecological inference technique, like other random coefficient models, uses conditional means as estimates. Consequently, if the target of an EI estimate is relatively large, then the EI estimate errs on the low side. Similarly, EI estimates err on the high side when trying to estimate a quantity that is relatively low. We show that these error patterns lead to attenuation bias in second-stage regressions—that, in fact, second-stage slope estimates are almost always less than half as large in magnitude as they should be.

One can imagine, for instance, an EI-based second-stage regression that has King-based estimates of voter turnout rates by precinct on its left-hand side. If, say, one of the explanatory variables in the regression reflects a government policy that varies by precinct, then our attenuation bias result implies that second-stage estimates dramatically understate the policy's impact on voter turnout.

It is worth noting that this inconsistency is not alleviated by using weighted least squares in EI-R's second stage. Weighting can be useful in the presence of heteroskedastic disturbance terms, but issues such as heteroskedasticity are not relevant if regression coefficient estimates are inconsistent to begin with.

After showing that EI-R, in general, produces inconsistent slope estimates, we propose a remedy for this problem. That such a remedy can be found is somewhat surprising because it is usually impossible to fix regression inconsistency caused by measurement error. In general, if one has information on the extent to which measurement errors are correlated with regressors, then it is possible only to place bounds on underlying population parameters (Bekker et al. 1987).

Fortunately, however, in some cases EI-R is an exception to this rule. We show that, although errors in individual EI estimates cannot be corrected, it is nonetheless often possible to estimate and then eliminate the errors' combined effect on second-stage regression estimates. This can be done because King's technique, under the assumption that it is

valid, generates consistent estimates of the confounding factors that cause second-stage inconsistency.

The remainder of this article proceeds as follows. In Section 2, we provide a brief example of problems with EI-R by modeling Broward County, Florida, voting behavior in the 2000 general election. In Section 3, we describe King's ecological inference technique in detail and explain why EI estimates are contaminated with error, that is, we explain why there is first-stage measurement error that precedes a second-stage regression. Section 4 provides a simple consistency condition for EI-R, and in Section 5 we demonstrate that this condition is almost never satisfied. In Section 6, we present evidence that a key parameterization used in our consistency condition works extremely well, and in Section 7 we offer a fix for second-stage inconsistency and describe circumstances in which the fix seems to work. In Section 8, we make constructive suggestions for researchers who want to avoid using EI-R, and, finally, Section 9 presents conclusions.

2 Failure of EI-R Applied to Broward County, Florida

We begin our study of EI-R with an example based on presidential and senatorial election returns from Broward County, Florida. Ignoring absentee voting, there were 606 Broward precincts in the 2000 general election. We have the complete record of ballot images for Broward County and thus can calculate precinct-level straight-ticket and ticket-splitting rates. We focus in this section on straight-ticket voting rates among voters who supported Democratic U.S. Senate candidate Bill Nelson. In a given precinct, the straight-ticket voting rate is the fraction of Nelson voters in a precinct who also voted for the 2000 Democratic candidate for president, Al Gore.

Ordinarily, of course, ballot-level data are not available to researchers and precinct-level straight-ticket voting rates can only be estimated from precinct-level election returns. In light of this, we use King's ecological inference technique to estimate precinct-level straight-ticket voting rates for Broward County and then run a regression of straight-ticket voting estimates (EI estimates, in this instance) on precinct-level demographic characteristics of registered voters; this two-step procedure is how a researcher would use EI-R to study Broward County voting.¹ However, because we know actual straight-ticket voting rates for Broward County, we can compare our EI-R results with results generated by a regression of *true* straight-ticket voting rates on precinct-level registration demographics. Table 1 gives the results of this comparison.²

Before discussing the implications of Table 1, two comments are in order. First, the unit of analysis in the table's two regressions is a precinct; hence, the variable Fraction Democratic is the fraction of registered voters in a Broward County precinct who self-identified with the Democratic Party. Second, the EI-R coefficients are estimated using a weighting approach proposed, and also caveatted, by King (1997, p. 290).

If EI-R performed well on the Broward County data, then we would expect to see coefficient ratios in the fourth column in Table 1 that are close to 1—indicating that the coefficient estimates based on estimated straight-ticket voting rates are similar to those based on true straight-ticket voting rates. We would also expect to see a high degree of similarity in significance levels (i.e., very few significance switches) across columns 2 and 3 in the

¹We used EzI version 2.3, downloaded from <http://gking.harvard.edu> on August 7, 2001. The results reported in Table 1 use EzI's default priors. We found similar results after estimating second-stage regressions based on first-stage EI estimates produced without prior distributions.

²Registration data used in Table 1 were obtained from the Broward County Supervisor of Elections.

Table 1 Comparison of two regressions from Broward County, Florida

<i>Variable</i>	<i>True coefficient</i>	<i>EI-R coefficient</i>	<i>Ratio of EI-R/True</i>	<i>Significance switch</i>
Constant	0.752* (0.043)	0.958* (0.020)	1.27	No
Fraction Democratic	0.279* (0.016)	0.127* (0.013)	0.43	No
Fraction third party	0.186 (0.126)	−0.026 (0.034)	−0.43	No
Fraction black	−0.017 (0.021)	−0.043* (0.00840)	2.96	Yes
Fraction Hispanic	−0.013 (0.034)	−0.054* (0.023)	4.66	Yes
Fraction male	0.014 (0.032)	−0.034 (0.028)	−3.08	No
Fraction over 56 years of age	−0.00344 (0.025)	−0.045* (0.00885)	16.26	Yes

Note. Heteroskedastic-consistent standard errors in parentheses. * $p < .05$. Significance switches calculated at the .05 level.

table—indicating that the two regressions produced similar statistical inferences regarding the key factors that affect straight-ticket voting rates.

However, these two expectations are not borne out. The EI-R estimate of the impact of Fraction Democratic on precinct-level straight-ticket voting is approximately one-half the coefficient based on true straight-ticket voting rates. Also, based on associated standard errors, confidence intervals for these two estimates of Fraction Democratic do not overlap. Moreover, the EI-R estimates for Fractions Black, Hispanic, and Over 56 Years of Age are all too large, and the estimates for Fraction Third Party and Fraction Male have incorrect signs.

Arguably, the problems described previously might not be so troublesome if the significance levels were identical for each true slope estimate in Table 1 and its corresponding EI-R estimate. However, these significance levels differ as summarized in the rightmost column in the table. For instance, according to the regression that uses true straight-ticket voting rates on its left-hand side, the Fraction Black in a precinct has no impact on straight-ticket voting (or, more precisely, because $t = -0.812$ there is no evidence to indicate that Fraction Black has an impact). The EI-R results imply, however, that Fraction Black has a very significant impact on straight-ticket voting ($t = -5.17$). Clearly, there should not be more information in the EI-R regression (recall that this regression has an estimated dependent variable) than in the regression that uses true straight-ticket voting rates.

At the very least, Table 1 should draw attention to the possibility that EI-R might be more troublesome than is commonly perceived. Moreover, it is important to recognize that the only difference between the two regressions in the table is that the first uses true straight-ticket voting rates on its left-hand side and the second uses estimates of these rates. Thus, the differences between column 2 and column 3 in the table *must* stem from first-stage errors in the King-based estimates of Broward County straight-ticket voting rates, that is, errors in what we have called EI estimates. In light of this, the next section describes in detail how EI estimates are generated and why they are contaminated with measurement error.

3 Generating EI Estimates for Use in a Second-Stage Analysis

For purposes of transparency between this article and existing work, when possible we adopt the notation used by King (1997, chap. 2). Suppose that for a given election and a collection of p precincts we have data on precinct i 's turnout rate $T_i \in [0, 1]$ and the percentage $X_i \in [0, 1]$ of precinct i 's voting age population that is black. King's technique uses T_i and X_i to estimate β_i^b , the fraction of blacks in precinct i that voted in the election, and β_i^w , the corresponding fraction of whites who voted. Note that β_i^b and β_i^w are disaggregated quantities, whereas T_i and X_i are aggregated.³

For each precinct i , the two unknown quantities β_i^b and β_i^w satisfy an accounting identity

$$\beta_i^w = \frac{T_i}{1 - X_i} - \beta_i^b \frac{X_i}{1 - X_i}. \quad (1)$$

Equation (1) implies that, given X_i and T_i , feasible values of β_i^b and β_i^w fall on a downward-sloping line, what King calls a *tomography line*.

A full set of assumptions and details regarding how King's ecological inference technique produces estimates of β_i^b and β_i^w based on X_i and T_i , $i = 1, \dots, p$, can be found in Chapters 6–8 of his book. There is an ongoing debate about the merits of King's technique and the importance of the assumptions that support it; see Tam (1998), Freedman et al. (1998, 1999), King (1999), Cho and Gaines (2000), and McCue (2001). In this article, however, we take as given the technique's validity and its assumptions.

In brief, the standard version of King's technique assumes that (β_i^b, β_i^w) have a truncated bivariate normal distribution on the unit square, that β_i^b and β_i^w are uncorrelated with X_i , and that, conditional on X_i , T_i and T_j are independent for $i \neq j$. The second assumption implies no aggregation bias and the third implies no spatial autocorrelation. King's technique estimates the five parameters—two means, two variances, and a correlation—of the truncated normal and then uses these parameter estimates to produce point estimates of β_i^b and β_i^w , $i = 1, \dots, p$. In King's notation, the parameters of the truncated normal are collected in a vector $\psi = \{\beta^b, \beta^w, \sigma_b, \sigma_w, \rho\}$, where β^b is the mean of β_i^b , σ_b^2 the variance of β_i^b , and so forth.⁴

Related to ψ is a set of five parameters associated with an untruncated bivariate normal distribution. These parameters are collected in a vector $\check{\psi} = \{\check{\beta}^b, \check{\beta}^w, \check{\sigma}_b, \check{\sigma}_w, \check{\rho}\}$. The precise relationship between ψ and $\check{\psi}$ is explained on pp. 102–106 of King's text, but for our purposes it is sufficient to note that $\check{\psi}$ implicitly defines ψ . Implementing King's technique on an ecological data set X_i and T_i , $i = 1, \dots, p$, leads to an estimate $\check{\psi}$ of ψ .

King defines the point estimate $\hat{\beta}_i^b$ of the disaggregated quantity β_i^b to be the mean of the truncated univariate normal distribution parameterized by $\hat{\psi}$ and conditional on X_i . That is, $\hat{\beta}_i^b$ is what we call an *EI estimate*. Appendix C of King's (1997) book describes the moments of the truncated univariate normal, and hence a formula for $\hat{\beta}_i^b$, but McCue (2001) argues that these formulas are wrong. McCue then proposes his own estimator of β_i^b . For the purposes of EI-R, the problems identified in this article are present regardless of whether one uses King EI estimates or McCue's proposed replacements.

³Of course, X_i , T_i , β_i^b , and β_i^w all refer to precinct-level quantities and are therefore aggregated. We say, however, that β_i^b and β_i^w are disaggregated insofar as they are less aggregated and contain more information than X_i and T_i .

⁴Because (β_i^b, β_i^w) are assumed to have a truncated bivariate normal distribution, one of X_i and T_i must be endogenous to this distribution. We assume, because $T_i = \beta_i^b X_i + \beta_i^w (1 - X_i)$ from Eq. (1), that X_i is exogenous and T_i endogenous.

Under the assumptions that support King's ecological inference technique, likelihood theory implies that $\check{\psi} \xrightarrow{\text{Pr}} \psi$ in the number of precincts p , where $\xrightarrow{\text{Pr}}$ denotes convergence in probability. It is important to recognize—indeed, it is practically obvious—that consistency of $\check{\psi}$ does not imply that the point estimate $\hat{\beta}_i^b$ approaches its true target of interest β_i^b as p tends to infinity (and this is unrelated to McCue's claims about problems in King's calculation of $\hat{\beta}_i^b$). In particular, a typical ecological data set with p observations includes $2p$ unknowns, namely, the values of β_i^b and β_i^w , $i = 1, \dots, p$. Consequently, ecological data sets have the feature that, as the number of observations gets large, so does the number of unknowns. We should not expect any statistical technique to produce consistent estimates of $2p$ disaggregated quantities based on p aggregated observations. The fact that King's technique cannot achieve this simply reflects the difficulty of the ecological inference problem. Nonetheless, and despite King's warning (1997, p. 111, emphasis in original) that “perfect knowledge of $\check{\psi}$ does not imply perfect knowledge of the precinct-level parameters of interest β_i^b and β_i^w After all, larger numbers of precincts do not eliminate the ecological inference problem,” researchers routinely treat the point estimate $\hat{\beta}_i^b$ as if it were consistent for β_i^b .

Just as $\hat{\beta}_i^b$ does not tend to β_i^b , $\hat{\beta}_i^w$ does not approach β_i^w as p tends to infinity. Thus, we define θ_i^b as the measurement error induced by using $\hat{\beta}_i^b$ to estimate β_i^b , and we similarly define θ_i^w :

$$\theta_i^b = \hat{\beta}_i^b - \beta_i^b, \quad (2)$$

$$\theta_i^w = \hat{\beta}_i^w - \beta_i^w. \quad (3)$$

As shown next, it is the fact that these two error terms are correlated with β_i^b and β_i^w , respectively, that causes problems for EI-based second-stage regressions. Notably, our comments about θ_i^b and θ_i^w apply even when all assumptions behind King's ecological inference technique are satisfied; we are not, that is, drawing a conclusion about EI estimates based on criticizing one of King's assumptions.

For the remainder of this article we focus primarily on the true disaggregated quantities β_i^b , their King-based estimates $\hat{\beta}_i^b$, and the resulting measurement errors θ_i^b . Our arguments apply in a symmetric fashion to β_i^w , $\hat{\beta}_i^w$, and θ_i^w .

4 EI Estimation Error and Second-Stage Consistency

We now offer a parameterization of EI measurement error θ_i^b so that we can investigate the general theoretical consequences of errors in EI estimates for the coefficients produced by EI-R.

4.1 Parameterizing EI Estimation Error

From Eq. (2) it follows that $\hat{\beta}_i^b = \beta_i^b + \theta_i^b$. Because the definition of the EI estimate $\hat{\beta}_i^b$ depends on the normal integral (recall that, according to King, $\hat{\beta}_i^b$ is the mean of a univariate truncated normal random variable) as well as on X_i , and because King's ecological inference technique does not produce a closed-form solution for the vector $\check{\psi}$, it is impossible to specify analytically the precise relationship between β_i^b and $\hat{\beta}_i^b$. Later in this article, we present evidence that EI measurement error (θ_i^b) is correlated with true underlying disaggregated quantities (β_i^b), and hence we approximate EI measurement error as follows:

$$\theta_i^b = \delta_0^b + \delta_1^b \beta_i^b + \zeta_i, \quad (4)$$

where δ_0^b and δ_1^b are scalar constants and ζ_i is a mean-zero disturbance term, assumed to be uncorrelated with β_i^b , with variance σ_ζ^2 . We assume that $\delta_1^b \neq -1$, and hence

$$\beta_i^b = \frac{\hat{\beta}_i^b - \delta_0^b - \zeta_i}{1 + \delta_1^b}. \quad (5)$$

If in fact $\delta_1^b = -1$, then Eqs. (2) and (4) imply that $\hat{\beta}_i^b = \delta_0^b + \zeta_i$, that is, the EI estimate $\hat{\beta}_i^b$ is unrelated to its intended target β_i^b . We rule out this pathological situation.

The parameter δ_1^b in Eq. (4) describes the extent to which EI measurement error θ_i^b is correlated with the associated disaggregated quantity β_i^b . Namely, regression theory implies that

$$\delta_1^b = \frac{\text{Cov}(\beta_i^b, \theta_i^b)}{\text{Var}(\beta_i^b)}, \quad (6)$$

and from this it follows that $\delta_1^b = 0$ if and only if $\text{Cov}(\beta_i^b, \theta_i^b) = 0$.

Equation (4) represents an approximation of EI estimation error. It is certainly true that this error, which must lie between -1 and $+1$, cannot be linear in β_i^b . Nonetheless, we adopt the linear error parameterization in Eq. (4) for four reasons. First, as shown immediately below, it leads to a straightforward characterization of the inconsistency that affects EI-R slope estimates. Second, we provide evidence later in the article that the parameterization works very well. Third, the parameterization leads to a fix for second-stage inconsistency. Fourth, and most important, the key element in the parameterization is whether $\text{Cov}(\beta_i^b, \theta_i^b)$ is or is not zero, and note that δ_1^b is simply a scaled version of $\text{Cov}(\beta_i^b, \theta_i^b)$. If $\text{Cov}(\beta_i^b, \theta_i^b) \neq 0$, then the estimates from an EI-based second-stage regression will be inconsistent, regardless of the validity of our linear error parameterization.

4.2 Implications of the Error Parameterization

Based on the measurement error parameterization of Eq. (4), we now determine conditions under which EI-based second-stage regressions are consistent. Let Z_i be a k vector of exogenous, precinct-level variables where $k + 1 < p$. A researcher studying factors that influence voter turnout across a set of p precincts might want to use least squares to estimate

$$\beta_i^b = \alpha + \gamma' Z_i + v_i, \quad (7)$$

where α is a scalar, γ is a k vector of unknown parameters, and v_i is a mean-zero disturbance term assumed to be uncorrelated with the elements of Z_i . Let γ_j denote the j th element of the parameter vector γ , $j = 1, \dots, k$.

Because β_i^b is unknown, King (1997, pp. 170, 279, 290) suggests that researchers carry out his inference technique, calculate $\hat{\beta}_i^b$, and then estimate

$$\hat{\beta}_i^b = \alpha + \gamma' Z_i + v_i. \quad (8)$$

Let $\hat{\alpha}$ and $\hat{\gamma}$ be the least-squares estimates of α and γ , respectively, based on p observations and Eq. (8). The estimate of the j th element of γ is $\hat{\gamma}_j$, $j = 1, \dots, k$.

It is logical to inquire about the types of distributions on v_i in second-stage Eqs. (7) and (8) that are compatible with the first-stage King ecological inference technique. It is similarly

logical to consider whether the assumptions that support the second-stage equations are compatible with the assumptions necessary for first-stage implementation of King's technique (e.g., Herron and Shotts 2002). At present, we take at face value the two-stage procedure that King advocates and others have used and assume that, measurement error notwithstanding, the two stages are logically consistent.

The following proposition details the consequences of EI estimation error for the least-squares slope estimate $\hat{\gamma}$ based on a four-equation model: Equation (7) describes the true relationship between Z_i and β_i^b ; Eq. (8) is the regression model used to estimate this relationship; Eq. (2) defines the measurement error that distinguishes Eq. (7) from Eq. (8); and, finally, Eq. (4) describes a model of the measurement error.

Proposition 1. *The least-squares slope estimate $\hat{\gamma}$ behaves as follows:*

$$\hat{\gamma} \xrightarrow{\text{Pr}} \gamma(1 + \delta_1^b). \quad (9)$$

Proposition 1, which follows from a simple algebraic substitution, shows via Eq. (9) that $\hat{\gamma}_j$ is consistent for γ_j when $\gamma_j = 0$, the special case in which there is no relationship between the j th element of the exogenous vector Z_i and the true dependent variable β_i^b . Furthermore, if $\gamma_j = 0$, then $\hat{\gamma}_j$ is consistent regardless of δ_1^b , that is, regardless of the extent to which EI measurement error θ_i^b is correlated with the true dependent variable β_i^b from Eq. (7). We provide additional intuition on the $\gamma = 0$ case later, but first discuss what happens when $\gamma \neq 0$.

In the case $\gamma_j \neq 0$, Eq. (9) implies that $\hat{\gamma}_j$ is not consistent unless $\delta_1^b = 0$. Thus, we refer to $\delta_1^b = 0$ as *the second-stage consistency condition*. Recall that we noted previously that $\delta_1^b = 0$ if and only if $\text{Cov}(\beta_i^b, \theta_i^b) = 0$, that is, there is zero covariance between the errors in EI estimates and the true values. Thus, if $\text{Cov}(\beta_i^b, \theta_i^b) \neq 0$, meaning that the EI-based measurement error θ_i^b is correlated with the disaggregated quantity β_i^b , then it follows that $\hat{\gamma}_j$ is inconsistent. What can we say about $\hat{\gamma}_j$ in this case? Suppose that the covariance between θ_i^b and β_i^b is negative and that, more specifically, δ_1^b lies between -1 and zero (in Section 4, we show that this is to be expected). Then, $1 + \delta_1^b$ is between zero and 1 and hence $\hat{\gamma}_j$ is biased to zero. That is, $\hat{\gamma}_j$ suffers from attenuation bias. Note, however, that this does not mean that significance tests based on $\hat{\gamma}_j$ are excessively conservative because inconsistent slope estimates of γ_j yield inconsistent estimates of regression variance and thus incorrect t statistics.

Incorrect t statistics are themselves problematic insofar as they cannot be used for statistical inference. Moreover, in the empirical domains studied with EI-based second-stage regressions, researchers often care about the actual values of slope coefficients. Suppose that we were to analyze white voters' willingness to vote for African-American congressional candidates. One way of doing this while controlling for candidate partisanship would be to use King's ecological inference technique to estimate the percentage of whites in different districts who voted for Democrats and then use these estimates as left-hand side variables in a second-stage regression. By including on the right-hand side of such a regression a dummy variable for race of the Democratic candidate, we could in theory estimate the effect of a candidate's race on his or her level of support among white voters.

Consider a hypothetical finding that, *ceteris paribus*, black Democratic candidates receive 20% fewer white votes than white Democratic candidates. Given the biases identified in Proposition 1, such a finding would understate the true penalty that white voters impose on black candidates. Based on our Section 5 analysis of real data sets from King's book,

the magnitude of this attenuation bias could be quite substantial, and the true magnitude of the penalty could easily be 2.5 times larger than the estimated penalty.

Returning to the special case of $\gamma = 0$ (i.e., $\gamma_j = 0$, $j = 1, \dots, k$), recall that in this case, $\hat{\gamma}$ is consistent regardless of the magnitude of δ_1^b . When $\gamma = 0$, then there is no relationship between Z_i and β_i^b ; furthermore, when there is no relationship between Z_i and β_i^b , then there is no relationship between Z_i and θ_i^b . Therefore, when $\gamma = 0$, the left-hand side measurement error that affects Eq. (8) is unrelated to any exogenous variables and simply gets absorbed into the regression disturbance.

This linkage between γ and the relationship between Z_i and θ_i^b indicates that, in general, left-hand side measurement error in linear regression models is a problem only when the error is correlated with right-hand side variables (this occurs when there is at least one element of γ that is nonzero). Ultimately, then, the source of second-stage inconsistency is the correlation between second-stage exogenous variables (Z_i) and first-stage measurement error (θ_i^b). This suggests that one approach a researcher might pursue is breaking this correlation by estimating an EI-based second-stage regression in such a way that Z_i is uncorrelated with θ_i^b . We discuss this in Section 8.

Finally, Proposition 1 shows implicitly why, when $\delta_1^b \neq 0$, the practice of using weighted least squares in an EI-based second-stage regression does not yield consistent slope estimates. Weighted least squares is an appropriate estimation technique when the ordinary least-squares estimates from a linear model are consistent yet have incorrect estimated standard errors as a result of the presence of a heteroskedastic disturbance term. In contrast, $\delta_1^b \neq 0$ implies that the least-squares slope estimates of Eq. (8) are inconsistent, and therefore matters such as heteroskedasticity are not relevant.

As an aside, the discussion by Lewis (2000) of weighting in second-stage regressions is predicated on $\text{Cov}(\beta_i^b, \theta_i^b) = 0$, that is, $\delta_1^b \neq 0$. If $\delta_1^b \neq 0$, then Lewis's discussion does not apply to EI-R. Similarly, King (1997, p. 290) proposes a two-step, second-stage approach in which the first step is a homoskedastic regression such as Eq. (8). Because this first regression yields inconsistent results, the second step, whose output is based on the first, is also inconsistent.

5 Failure of the Second-Stage Consistency Condition

We now consider whether the second-stage consistency condition generally holds in ecological data sets. It is straightforward to confirm that the condition fails in the Broward County voting data used to introduce this article's study of EI-R; for Broward County, $\delta_1^b = -0.484$ ($t = -29.5$), and the correlation between β_i^b and θ_i^b is -0.77 . To investigate the second-stage consistency condition more generally, in this section we adopt all assumptions behind King's ecological inference technique. We nonetheless find that, except in unusual circumstances, $\text{Cov}(\beta_i^b, \theta_i^b)$ is negative and thus $\delta_1^b < 0$. We first explain the intuition behind this result, then demonstrate that it occurs in a wide range of simulated data, and, finally, confirm our findings by examining a real ecological data set.

5.1 Intuition on the Consistency Condition

To see why errors in EI estimates are negatively correlated with true, underlying disaggregated quantities, consider Fig. 1. This figure contains a downward-sloping tomography line that characterizes a single precinct, given its values of X_i and T_i . The line captures all values of β_i^b and β_i^w that are possible based on the accounting identity in Eq. (1).

Once King's technique operates on an ecological data set, X_i and T_i , $i = 1, \dots, p$, and produces an estimate $\hat{\psi}$ of the vector ψ , the tomography line for each precinct i has a

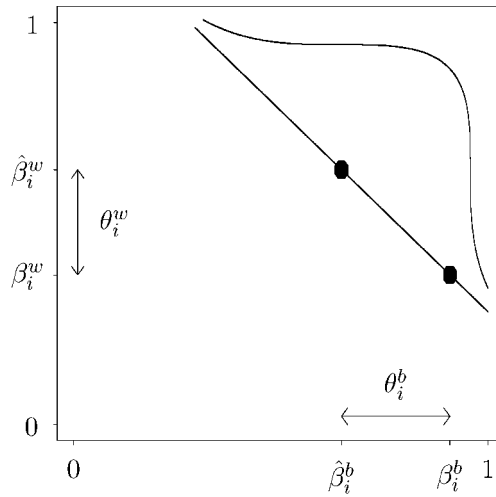


Fig. 1 A single tomography line in $\beta_i^b \times \beta_i^w$ space.

distribution defined on it in which this distribution is based on the assumption that (β_i^b, β_i^w) have a truncated bivariate normal distribution. In line with King's claims, the curve on top of the tomography line in Fig. 1 illustrates a truncated univariate normal conditional on the precinct's particular values of X_i and T_i . However, McCue (2001) claims that the distribution in Fig. 1 is not actually a truncated univariate normal, and that, in fact, no one has yet characterized the distribution. From our perspective, whether or not the distribution is truncated univariate normal is irrelevant. What is important, as is shown later, is that the mean of the distribution on the tomography line is used to estimate $\hat{\beta}_i^b$ and that this mean is found approximately under the middle of the curve in Fig. 1.

Suppose the precinct described in Fig. 1 had a rather large value of β_i^b , a value located in the far-right tail of the pictured distribution. In this case, the measurement error θ_i^b defined in Eq. (2) is negative because $\theta_i^b = \hat{\beta}_i^b - \beta_i^b$ and $\hat{\beta}_i^b < \beta_i^b$.

Figure 1 illustrates another point that is very important in light of Proposition 1. Consider the set of possible values of β_i^b that lie on the tomography line. For precincts with β_i^b larger than average for this group, the error induced by using $\hat{\beta}_i^b$ to estimate β_i^b is negative. For precincts with β_i^b smaller than average, however, the error is positive. This form of regression to the mean implies that, in general, for precincts characterized by this tomography line there should be a negative covariance between β_i^b and θ_i^b .

Because Fig. 1 contains only a single tomography line and the covariance between β_i^b and θ_i^b depends on p different tomography lines, the figure should not be understood as proving that $\text{Cov}(\beta_i^b, \theta_i^b) \neq 0$. Therefore, our examination of a single tomography line is necessarily limited, and the next section presents simulations that examine large numbers of tomography lines.

5.2 Simulation Evidence on the Consistency Condition

We now explore with simulations the extent to which β_i^b is correlated with θ_i^b , that is, the extent to which $\text{Cov}(\beta_i^b, \theta_i^b) \neq 0$. In each simulation, we examine a large number of precincts for which β_i^b and $\hat{\beta}_i^b$ are known and estimate a regression based on Eq. (4). We then calculate a t statistic for δ_1^b and check to see if it is large in magnitude. If it is, we

conclude that the arrangement of precincts being simulated was one in which EI estimation error leads to second-stage inconsistency.

The simulations proceed as follows. First, we specify the five elements of the vector $\check{\psi}$ and calculate the associated vector ψ . The latter is derived as follows: We draw 200,000 bivariate normal random variables based on the means, variances, and correlation in $\check{\psi}$, discard all values outside of the unit square, and then use the sample means, sample variances, and sample covariances of the remaining draws to specify ψ . We select without replacement $p = 20,000$ of the remaining values to represent true (β_i^b, β_i^w) pairs. This precisely mimics King's assumption that (β_i^b, β_i^w) have a truncated bivariate normal distribution.

We next draw p values of X_i from a uniform distribution on $[0, 1]$ and infer the requisite value of T_i , $i = 1, \dots, p$, based on the accounting identity in Eq. (1). Then we assume that $\hat{\psi}$, which in theory is based on implementing King's ecological inference technique on T_i and X_i , $i = 1, \dots, p$, exactly recovers $\check{\psi}$. This is a conservative assumption, as in reality the technique implies that $\hat{\psi} \xrightarrow{\text{Pr}} \check{\psi}$ rather than $\hat{\psi} = \check{\psi}$. Nonetheless, we assume that $\hat{\psi} = \check{\psi}$ so that the conclusions of our simulations are not affected by first-stage errors in the estimation of $\check{\psi}$. Next, based on $\hat{\beta}_i^b$, we calculate θ_i^b from Eq. (2) and then estimate the regression in Eq. (4). Finally, we record the δ_1^b t statistic.

Our simulations allow for the following configurations of $\check{\psi}$:

$$\begin{aligned}(\check{\beta}^b, \check{\beta}^w) &\in \{(0.5, 0.5), (0.5, 0.85), (0.85, 0.85), (0.85, 0.25)\}, \\(\check{\sigma}_b, \check{\sigma}_w) &\in \{(0.173, 0.173), (0.141, 0.2), (0.224, 0.1)\}, \\ \check{\rho} &\in \{0, -0.4, -0.9, 0.6\}.\end{aligned}$$

This leads to a total of 48 different parameterizations of the vector $\check{\psi}$ and, accordingly, 48 different t statistics for δ_1^b .⁵

Of the 48 t statistics for δ_1^b generated by our simulations, none was smaller in magnitude than 65. All the estimated values of δ_1^b were between -1 and zero.

For example, consider the set of simulations for which $(\check{\beta}^b, \check{\beta}^w) = (0.5, 0.5)$, $(\check{\sigma}_b, \check{\sigma}_w) = (0.173, 0.173)$, and $\check{\rho} = 0$. These values lead to a t statistic for δ_1^b of -139 , and the point estimate of δ_1^b is -0.493 . We thus reject that δ_1^b is zero, and it follows that EI-based second-stage regressions based on these five parameter values are inconsistent. In particular, Proposition 1 implies that a second-stage regression that used corresponding EI estimates as left-hand-side variables would produce slope estimates off by a factor of $1/(1 - 0.493) \approx 1.97$.

5.3 The Consistency Condition in a Real Ecological Data Set

Our simulations show that δ_1^b , and hence the covariance between β_i^b and θ_i^b , are, in general, negative. This is important in light of the theoretical results in Proposition 1 that link δ_1^b to second-stage inconsistency. Nonetheless, all simulated data sets suffer from the potential problem that they are artificial and thus might not be representative of typical situations encountered by empirical researchers. Therefore, we now consider King's data on voter turnout in the state of Louisiana.

King (1997, p. 23) displays a plot of β_i^b versus $\hat{\beta}_i^b$ for a collection of 3262 precincts in Louisiana in 1990. For the Louisiana data set, β_i^b is the true African-American turnout in Louisiana precinct i and $\hat{\beta}_i^b$ is the corresponding point estimate generated by applying

⁵All statistics and software code are available at the *Political Analysis* Web site.

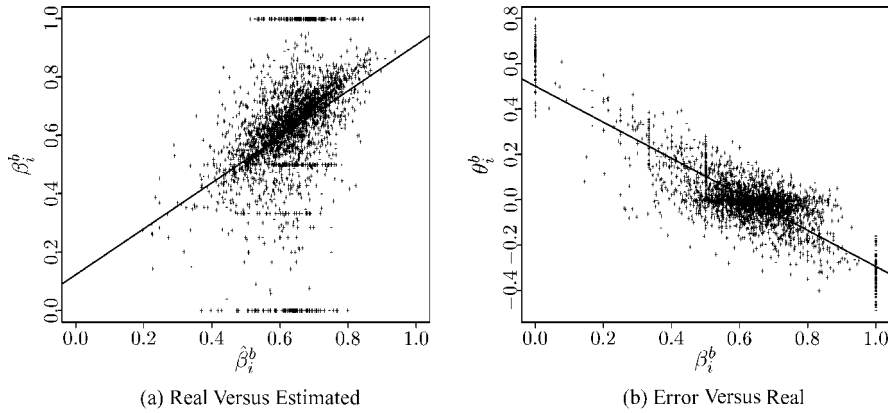


Fig. 2 EI estimates of African-American turnout in Louisiana in 1990.

King's ecological inference technique to observed values of black population X_i and total turnout T_i .⁶

Figure 2 presents two plots based on the Louisiana data set. In Fig. 2a, β_i^b is plotted against the EI estimate $\hat{\beta}_i^b$, and the result is a graph very similar to that on King's (1997) page 23.⁷ In Fig. 2b, we plot the EI estimation error θ_i^b on the vertical axis against the true African-American turnout β_i^b on the horizontal axis.

Both plots in Fig. 2 include superimposed regression lines. In Fig. 2a, the slope of the associated regression line is 0.79, with an estimated standard error of 0.033. Because the 95% confidence interval for the slope is (0.86, 0.72), we can reject at conventional confidence levels the hypothesis that the slope is 1.⁸

As King points out, the general form of the "Real Versus Estimated" plot is visually pleasing. Figure 2b, however, displays a different perspective on the relationship between the Louisiana EI estimates and the actual turnout rates. Namely, this plot shows that the EI-based measurement error is negatively correlated with the true African-American turnout levels, that is, high levels of β_i^b are associated with low θ_i^b . This is an important result, because our earlier evidence on the covariance between β_i^b and θ_i^b was based on examples and simulated data. We have now examined a real data set, and we find a negative correlation here as well.

In particular, the regression line in Fig. 2b, in which θ_i^b is regressed on β_i^b and a constant, has a slope or δ_1^b value of -0.795 ($t = -91.6$). We can calculate the attenuation bias implied by the linear error parameterization in a second-stage regression that uses the Louisiana $\hat{\beta}_i^b$ values as dependent variables. Specifically, from Proposition 1 we see that such a second-stage regression would understate all slope values by a factor of $1/(1 - 0.795) \approx 4.88$.⁹

⁶Our analysis of the Louisiana data set relies entirely on data downloaded from <http://gking.harvard.edu>. Our EI estimates $\hat{\beta}_i^b$ are averages of 100 random draws from a set of univariate truncated normal distributions. All such draws are provided by King at his Web site, and we simply averaged them by precinct.

⁷It appears that, in King's plot, the Louisiana precincts for which the African-American turnout was zero or complete are not visible as a result of being coincident with lower or upper horizontal axes.

⁸Our regressions ignore precincts in which $X_i = 0$ (no black residents) or $X_i = 1$ (no white residents). These precincts are uninteresting because they pose no ecological inference problem.

⁹We found results that were similar, although not necessarily as extreme, when analyzing other King (1997) data sets: southern counties registration (chap. 10), Kentucky registration (chap. 12), Fulton County transition rates (chap. 13), and black literacy in 1910 (chap. 13).

6 Evaluation of the Linear Error Parameterization

A virtue of our linear error parameterization is that it leads to a straightforward characterization of the second-stage inconsistency problem. Also, the parameterization is not binding in the sense that it sheds light on whether $\text{Cov}(\beta_i^b, \theta_i^b) = 0$ regardless of whether the parameterization is valid. Nonetheless, we have not yet offered any evidence that the parameterization provides a good approximation of the EI estimation error. We now address this matter.

Our test for parameterization usefulness is based on estimating a bivariate EI-based second-stage regression. Thus, following Eq. (8), suppose that we estimate $\hat{\beta}_i^b = \alpha + \gamma Z_i + v_i$. Assuming that our linear error parameterization is valid, Proposition 1 shows that the resulting estimate $\hat{\gamma}$ of γ is biased to zero and, in particular, tends in probability to $\gamma(1 + \delta_1^b)$.

Suppose that the linear parameterization is valid. Then, after estimating a second-stage regression and producing $\hat{\gamma}$, we could compute an adjusted estimate $\hat{\gamma}/(1 + \delta_1^b)$, and this should recover the true γ . In contrast, however, suppose that the true EI estimation error is far from being linear, yet we force our linear parameterization on it and produce a δ_1^b that is meaningless. In this case, the quotient $\hat{\gamma}/(1 + \delta_1^b)$ is not close to the real γ . In fact, one might conjecture that the adjusted estimate—the second-stage estimate $\hat{\gamma}$ divided by 1 plus the meaningless δ_1^b —is a worse estimate of γ than the regular, unadjusted, second-stage estimate.

This logic forms the basis of simulations designed to test the utility of our linear error parameterization. The simulations proceed as follows.

1. Specify a 5-vector $\check{\psi}$ (untruncated scale) and calculate the associated 5-vector ψ (truncated scale) using 50,000 draws from a bivariate normal parameterized by $\check{\psi}$.
2. Repeat the following loop 100 times.
 - a. Draw (β_i^b, β_i^w) , $i = 1, \dots, 5000$ ($p = 5000$), from the truncated bivariate normal distribution parameterized by ψ . Draw X_i from a uniform distribution on $[0, 1]$, and then calculate T_i from Eq. (1).
 - b. Assume that the first-stage ecological inference vector estimate $\hat{\psi}$ recovers $\check{\psi}$ exactly. Calculate $\hat{\beta}_i^b$ using the analytical mean of Johnson et al. (1994, Eq. 13.134), and then calculate θ_i^b from Eq. (2).¹⁰
 - c. Use Eq. (4) to estimate $\hat{\delta}_1^b$.
 - d. Define an exogenous covariate $Z_i = \beta_i^b + \tau_i$, where τ_i is a normal random variable with mean zero and standard deviation 0.5. This step preserves the fact that (β_i^b, β_i^w) has a truncated bivariate normal distribution. Had we specified Z_i first and then derived β_i^b from it, we would have violated an assumption essential to the validity of King's ecological inference technique.
 - e. Estimate with least squares $\beta_i^b = \alpha + \gamma_R Z_i + v_i$, where we call $\hat{\gamma}_R$ the real or true γ . Note that $\hat{\gamma}_R$ is based on true β_i^b values that in actual research would not be known.
 - f. Estimate $\hat{\beta}_i^b = \alpha + \gamma_U Z_i + v_i$, where γ_U is the unadjusted EI-R slope estimate. Let $\hat{\gamma}_U$ denote the resulting estimate of γ_U .
 - g. Finally, let $\hat{\gamma}_A = \hat{\gamma}_U/(1 + \hat{\delta}_1^b)$ denote the adjusted γ as in Proposition 1.

¹⁰The bounds for the mean of the truncated univariate normal come from T_i and X_i as described by King (1997, p. 79).

Table 2 Simulation results on the utility of the linear error parameterization

$\check{\psi}$	$ \hat{\gamma}_R - \hat{\gamma}_U $	$ \hat{\gamma}_R - \hat{\gamma}_A $	$(\hat{\gamma}_R - \hat{\gamma}_U)^2$	$(\hat{\gamma}_R - \hat{\gamma}_A)^2$	$\hat{\gamma}_U/\hat{\gamma}_R$	$\hat{\gamma}_A/\hat{\gamma}_R$
(0.5, 0.5, 0.1, 0.1, 0)	0.020	0.0022	3.9×10^{-4}	1.2×10^{-7}	0.50	0.99
(0.75, 0.5, 0.1, 0.1, 0)	0.018	0.0019	3.4×10^{-4}	4.1×10^{-8}	0.50	1.01
(0.75, 0.75, 0.1, 0.1, 0)	0.018	0.0021	3.4×10^{-4}	9.1×10^{-8}	0.50	1.00
(0.9, 0.9, 0.1, 0.1, 0)	0.012	0.0018	1.5×10^{-4}	1.4×10^{-7}	0.50	0.99
(0.5, 0.5, 0.32, 0.1, 0)	0.060	0.0027	3.6×10^{-3}	2.4×10^{-7}	0.68	1.00
(0.6, 0.6, 0.1, 0.32, 0)	0.10	0.0040	1.1×10^{-2}	1.5×10^{-8}	0.53	1.00
(0.9, 0.1, 0.32, 0.32, 0)	0.064	0.0036	4.1×10^{-3}	1.4×10^{-7}	0.55	1.00
(0.5, 0.5, 0.1, 0.1, 0.3)	0.015	0.0017	2.3×10^{-4}	1.0×10^{-8}	0.62	1.01

Note. All results are averages over 100 simulations.

The results in Table 2, which are based on 100 simulations, provide evidence in favor of our linear error parameterization. Columns 2 and 3 show that the average absolute error in $\hat{\gamma}_A$ is smaller than the corresponding error for $\hat{\gamma}_U$. Moreover, in every one of the 800 simulations the adjusted coefficient $\hat{\gamma}_A$ was closer to the true γ than the regular EI-R estimate $\hat{\gamma}_U$: The error in adjusted estimates is typically 10 times smaller (range of 6–24 times smaller) than the error in regular EI-R estimates. Columns 4 and 5 show that the average squared error in $\hat{\gamma}_A$ is much smaller than the average squared error in $\hat{\gamma}_U$. Finally, the last two columns show that the unadjusted second-stage coefficients suffer from attenuation bias, as predicted by our linear error parameterization, and that our correction via division solves this problem almost in its entirety. Note that the rightmost column in Table 2 is composed of numbers that are almost exactly equal to 1.

The evidence in Table 2 is noteworthy because it suggests that our linear error parameterization, which leads to straightforward results for second-stage consistency, is very useful. It is reasonable to be concerned that a linear parameterization of something that in reality is nonlinear could undermine our results. Nonetheless, the simulations described in this section show that our linear parameterization, despite being only an approximation, works well and leads to adjusted second-stage estimates that dominate regular second-stage estimates.

7 Estimation in the Face of Second-Stage Inconsistency

Our results thus far have identified a consistency condition for EI-R, and based on analysis of simulated and real ecological data sets, we showed that this condition is almost always violated. Hence, EI-R, except in highly unusual circumstances, produces inconsistent coefficient estimates. Furthermore, second-stage estimated standard errors are incorrect because inconsistent slope parameter estimates lead to inconsistent estimates of corresponding regression variance. Thus, as in the Broward County analysis, inference based on EI-R is likely to be misleading.

In light of this negative result, we now propose a fix for second-stage inconsistency. Before turning to this fix, however, we first reconcile our theoretical results on EI-R—recall that these results indicate that second-stage estimates suffer from attenuation bias—with the Broward County analysis that contained examples of coefficients that were too small in magnitude, too large, and even incorrectly signed. As will soon be apparent, this reconciliation effort has important implications regarding when our fix for EI-R inconsistency is applicable.

7.1 Second-Stage Attenuation Bias and When to Expect It

The reason that the EI-R coefficients for Broward County did not all suffer from attenuation bias is sampling variability. To the point, the most significant coefficient in the Broward County EI-based second-stage regression (Fraction Democratic) has a t statistic of 9.63 and suffers from attenuation bias (ratio of 0.43) that is quite close to what our linear error parameterization predicts (predicted ratio of 0.52 based on $\delta_1^b = -0.484$ for Broward County). Moreover, based on true Broward County straight-ticket voting rates, the Fraction Democratic slope estimate has a t statistic of almost 18: This is by far the most significant result in the regression. This is good news, in a sense; although the EI-R slope estimate for Fraction Democratic badly misses its true value, it nonetheless errs in a way that is predictable and straightforward to adjust based on our linear error parameterization.

However, the other coefficients from the Broward County EI-R application are much less well behaved in the sense of suffering from the 50% attenuation bias predicted by our error parameterization. An explanation for this stems from the fact that the t statistics for these slope coefficients, which range from approximately 0.8 to 5.2 in magnitude, are much smaller than the Fraction Democratic t statistic. This suggests the following point: Unless an EI-R slope estimate is extremely significant, even an adjusted version of the estimate can easily be far from the value that would be produced by a regression based on true dependent variable values. In other words, the Broward County example shows that EI-R estimates can suffer from unpredictable inaccuracies in finite samples, even if, asymptotically, they suffer from attenuation bias.

To assess the extent to which this conjecture is correct, we extend our simulations from Section 6. These simulations, designed to evaluate our linear error parameterization, were originally conducted with sample size $p = 5000$. We now repeat the simulations, letting p be drawn from a discrete uniform distribution on $[10, 5000]$. By varying p , we allow the significance of the unadjusted scalar slope estimate in the simulations to vary.

There are eight configurations of ψ in the Section 6 simulations (see Table 2 for details), and for each configuration we perform 2000 simulations in which p is drawn randomly. For the first configuration and its 2000 corresponding estimates of $\hat{\gamma}_U$, $\hat{\gamma}_R$, and $\hat{\gamma}_A$, we plot in Fig. 3 the ratio $(\hat{\gamma}_U/\hat{\gamma}_R)$ versus the magnitude of the unadjusted slope t statistic. We also plot the unadjusted squared error minus the adjusted squared error $((\hat{\gamma}_R - \hat{\gamma}_A)^2 - (\hat{\gamma}_R - \hat{\gamma}_U)^2)$ versus the magnitude of the unadjusted t statistic.

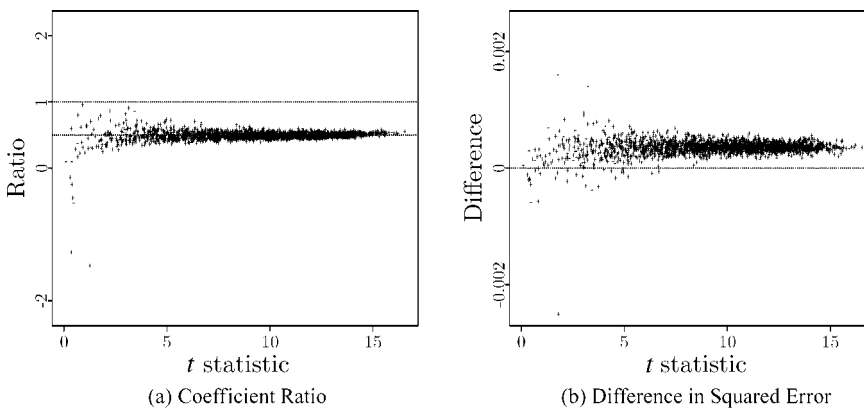


Fig. 3 Sampling variability and unadjusted t statistics.

The lower dotted line in Fig. 3a is at 0.5 and the upper dotted line is at 1. As noted in the top line in Table 2, 0.5 is approximately what the unadjusted-to-adjusted slope estimate ratio should be in the limit, according to underlying simulation parameters. The dotted line at one highlights estimated slope coefficients that suffer from augmentation bias, as opposed to attenuation bias.¹¹

Figure 3 has two implications. First, Fig. 3a shows that the coefficient ratio is less variable the larger the EI-R slope estimate t statistic. This decrease in variance is about the lower dotted line, and therefore Fig. 3a shows that the ratio becomes relatively more reliable when the EI-R t statistic is relatively large. However, for t statistics that are less than 4, the coefficient ratio can be quite unreliable (as, indeed, it was in the Broward County example). In other words, the predicted attenuation bias in second-stage slope estimates materializes when the EI-R second-stage t statistic is large.

Second, Fig. 3b shows that the adjusted coefficient estimate, as prescribed by the linear error parameterization, is almost always better in a squared error sense than the unadjusted estimate for unadjusted t statistics greater than approximately 8. One can see this by noting that the difference $(\hat{\gamma}_R - \hat{\gamma}_A)^2 - (\hat{\gamma}_R - \hat{\gamma}_U)^2$ is almost always positive once the unadjusted t statistic is relatively large.

7.2 Correcting for Second-Stage Attenuation Bias

Given the results in Fig. 3 as well as additional simulations based on the other seven $\tilde{\psi}$ configurations of Table 2, we recommend the following:

- When a slope estimate from an EI-based second-stage regression has a relatively small t statistic (less than 4), then the corresponding estimate cannot be trusted. The sign of the estimate may be wrong, the estimate may be too large in magnitude, and so forth. Second-stage estimates with unadjusted t statistics less than 4 should not be used to test hypotheses. This implies, among other things, that there is good reason to be quite skeptical of null results based on EI-R, such as Gay's (2001) finding that black turnout is unaffected by black congressional representation.

The Broward County example in Section 2 suggests that a t statistic cutoff value of 4 is not sufficiently conservative (note that the coefficient for fraction black is much too large despite the fact that the t statistic is greater than 5). We suspect that this is the case because our simulations match King's first-stage assumptions exactly. Real data, as exemplified by that from Broward County, do not do this and require greater conservatism in the t statistic cutoff.

- When the slope coefficient from an EI-based second-stage regression has a relatively large t statistic (conservatively, greater than 8), the estimate should be inflated by $1/(1 + \delta_1^b)$ as described in Section 6. We show later how to estimate δ_1^b .
- When the slope coefficient from an unadjusted EI-based regression has a medium-range t statistic (between 4 and 8), then it almost certainly suffers from attenuation bias. Our adjustment based on $1/(1 + \delta_1^b)$ is most likely better than the unadjusted estimate, but this is not guaranteed.

It is worth noting that the coefficient adjustment described previously is quite straightforward to implement, and the simulations in Section 6 show that it works very

¹¹ Several extreme outlier points were dropped from both Fig. 3a and Fig. 3b. Adding these points would not be qualitatively important, but it would lead to very stretched plots.

well, provided that the coefficient being adjusted (i.e., inflated) has a large t statistic. All that is needed to carry out the adjustment is a consistent estimate of δ_1^b . This estimate can be generated as follows:

1. Apply King's ecological inference technique to a collection of X_i and T_i , $i = 1, \dots, p$, to estimate $\tilde{\psi}$.
2. Draw p values of (β_i^b, β_i^w) from the truncated bivariate normal parameterized by $\tilde{\psi}$.
3. For each (β_i^b, β_i^w) pair, let $T_i = \beta_i^b X_i + \beta_i^w (1 - X_i)$ replace the observed T_i . Then, using the observed X_i , the new T_i just calculated, and the vector estimate $\hat{\psi}$ from Step 1, calculate the corresponding $\hat{\beta}_i^b$ using simulations described by King (1997) or the analytical approach of Johnson et al. (1994, Eq. 13.134). Finally, use Eq. (2) to derive the resulting measurement error θ_i^b .
4. Following Eq. (4), regress θ_i^b on β_i^b and a constant. As p gets large, the result is a consistent estimate $\hat{\delta}_1^b$ of δ_1^b .

To generate standard errors and conduct inference, one might be able to use a bootstrap that resampled over an ecological data set and computed many estimates of an adjusted coefficient. In practice, this would require numerous iterations of King's ecological inference technique.

It should not be forgotten that, for unadjusted EI-R estimates, both marginal effects *and* standard errors are wrong. Marginal effects are wrong because second-stage slope estimates are inconsistent, and standard errors are wrong because they are based on an inconsistent estimate of regression variance. In contrast, marginal effects based on adjusted second-stage estimates are consistent, even though it is not yet known how to produce associated standard errors in a computationally simple way.

7.3 Inaccuracy of Second-Stage Standard Errors

That second-stage standard errors are wrong asymptotically is obvious in light of second-stage inconsistency. Moreover, one can get a sense of this by calculating for Fig. 3 the fraction of the 95% confidence intervals for the true slope coefficient $\hat{\gamma}_R$ that actually contain this coefficient. In particular, of the 2000 confidence intervals based on $\hat{\gamma}_{\hat{U}}$ from the simulations in Fig. 3, only 2.4% contain $\hat{\gamma}_R$.

The presence of extremely inaccurate confidence intervals based on unadjusted EI-based second-stage regressions should not be considered surprising. After all, these intervals are based on slope estimates that are inconsistent and, in some cases, quite unpredictable. Nonetheless, this brief illustration of problems in standard errors is valuable insofar as some users of EI-R might be tempted to think that unadjusted second-stage estimates are better than adjusted estimates because the former have standard errors. It is indeed true that the former have easily computed standard errors; nonetheless, the standard errors are of very little value.

8 Avoiding EI-Based Second-Stage Regressions

We know that EI-R leads to inconsistent results. Moreover, our proposal for fixing second-stage inconsistency has some limitations because it cannot be applied to all second-stage slope estimates. We suspect that researchers who want to avoid the inconsistency of EI-R may seek different methods for analysis of ecological data. Here we present two ways to do this, as well as a way that appears attractive but actually is not.

8.1 King's Extended Model in Lieu of Second-Stage Regressions

A researcher considering using EI-R could instead use King's extended ecological inference model. In this model, exogenous covariates—those that might otherwise appear in a second-stage regression—are incorporated directly into the first stage. Thus, once covariates appear in the extended model, there is no second stage at all and no second-stage inconsistency problem.

The extended model highlights a fundamental problem with second-stage regressions. Namely, a second-stage regression conditions $\hat{\beta}_i^b$ on the exogenous Z_i vector, whereas King's standard ecological inference model produces $\hat{\beta}_i^b$ without conditioning in this way. This is an important point: Estimates of β_i^b conditional on Z_i generally differ from estimates of β_i^b not conditional on Z_i . This is, ultimately, what causes second-stage inconsistency. Although in this article we focus on the presence of a negative correlation between errors in EI estimates and true disaggregated quantities, we could have instead motivated our discussion by noting that a second-stage regression introduces additional conditioning variables, and thus induces inconsistency.

The extended model, it is worth pointing out, is barely identified (Rivers 1998). Implementations of it may be prone to numerical instabilities, and, at worst, an unidentified extended model might be treated as if it were identified. Moreover, such difficulties illustrate the complexity of using exogenous covariates in ecological inference. This complexity can be covered up by second-stage linear regressions that are too easily identified when unknown quantities (β_i^b) are treated as known ($\hat{\beta}_i^b$).

8.2 Additional Data in Lieu of Second-Stage Regressions

Another alternative is using data disaggregated one level below that which is truly of interest. For example, suppose that a researcher sought to study the factors that affect split-ticket voting at the congressional district level. A level below this is precinct, so the researcher could gather precinct-level data on voting behavior, apply King's ecological inference technique separately on each of 435 congressional districts, and thereby generate a consistent estimate of β^b for each district. This assumes, of course, that King's ecological inference technique is applicable to each district. In this example, β^b for a given district is the mean of the β_i^b values, where i indicates a precinct in the district of interest. In our earlier discussion of King's method, we had one value of β^b and many of β_i^b ; in this second alternative, we have 435 values of β^b , and for each of these there is a collection of β_i^b values. Then a second-stage regression could use the 435 estimated β^b values on its left-hand side.¹²

The advantage of this second alternative is that it uses estimates of β^b as a dependent variable. This contrasts with the EI-R's use of a dependent variable, $\hat{\beta}_i^b$, that is known to be inconsistent for its intended target. The drawback to the alternative, however, is that it requires a considerable investment in additional data.

8.3 A Nonsolution to Second-Stage Inconsistency

One intuitive, yet incorrect, way to ameliorate the second-stage inconsistency problem is to use what we call *averaged regressions*. This nonsolution proposes that, after applying King's ecological inference technique, researchers who want to use EI-R should instead

¹²In a related approach, Burden and Kimball (1997) compare precinct and district-level EI estimates.

draw one observation from each of the β_i^b truncated univariate normal distributions, estimate a regression with these p draws as dependent variables, and then repeat. According to the nonsolution, the average of the regression estimates should converge in probability to the desired regression coefficients.

The nonsolution fails, however, because the average of the regression estimates converges in probability to the same quantity that results when EI point estimates themselves are used in regressions. The proof of this is straightforward and follows from the fact that the average draw from a univariate truncated normal converges in probability to the mean of the distribution.

The problem with the nonsolution is that, for all possible β_i^b values on a given tomography line, the randomly drawn values that proxy for β_i^b are all drawn from the same pool. Thus, when β_i^b is large, the resulting draws tend to be too low, and when β_i^b is small, the resulting draws tend to be too high. On average, the mean of the resulting draws are $\hat{\beta}_i^b$, the EI point estimate itself. It follows that the averaged regressions technique induces the same inconsistency problems present in ordinary least-squares estimation of EI-based second-stage regressions.

9 Conclusion

We show that the practice of using point estimates produced by King's (1997) ecological inference technique as dependent variables in a second-stage regression leads, except in highly unusual circumstances, to inconsistent results. Nonetheless, this practice has been used by a variety of researchers and appears to be heading toward widespread popularity. We therefore have a striking situation: Researchers are publishing findings using a statistical technique that, even in the best of circumstances, does not work.

We show that second-stage inconsistency occurs even when all assumptions behind King's technique are satisfied. In particular, second-stage coefficient estimates are inconsistent because the point estimates produced by King's ecological inference technique are contaminated with error that, in almost all circumstances, is correlated with the disaggregated quantities the estimates seek to measure.

One possible reaction to the results presented here is to argue that, for researchers interested in the relationship between exogenous variables and disaggregated quantities, accepting inconsistent second-stage results is simply a cost that must be borne. Two reasons lead us to reject this argument. (1) The proposition that inconsistent second-stage estimates are better than no estimates at all cannot be scientifically adjudicated. (2) Second-stage estimates can be very inaccurate, and because what the standard errors from EI-based second-stage regressions really mean is unknown, inference based on them can be seriously misleading. Thus, accepting results from EI-based second-stage regressions lead to research results that are practically impossible to evaluate.

The guaranteed way to avoid second-stage inconsistency is to avoid second-stage regressions altogether. Researchers who want to pursue this should use King's extended ecological inference model, which includes covariates in its first stage.

For those who nonetheless want to use EI-R, we provide a fix for second-stage inconsistency and some guidelines that prescribe when this fix is likely to work. It is unknown, at the time of this article's writing, how to interpret second-stage estimates that arise in a context in which our fix is not applicable. In addition, it is unknown how to generate standard errors for adjusted second-stage estimates in a straightforward way.

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